# Detailed Classification of Fetal Congenital Heart Disease in This Study

## (Updated: January 2026)

The septal defects subgroup includes: atrial septal defect (ASD)—ostium primum/ostium secundum, coronary sinus type (unroofed coronary sinus syndrome), sinus venosus type (superior/inferior); ventricular septal defect (VSD)—outlet/inlet (restrictive/non-restrictive), membranous/perimembranous, muscular (single/multiple), doubly committed subarterial, subcristal/infracristal, subpulmonary (infundibular) variants, and all other constituent subtypes; atrioventricular septal defect (AVSD)—partial/intermediate/complete (balanced/unbalanced), as well as unbalanced AVSD with left- or right-dominant morphology.

The great vessels/arch configuration subgroup includes: transposition-related lesions—congenitally corrected transposition of the great arteries (ccTGA; IDD/IDL/SLD/SLL), complete transposition of the great arteries (d-TGA; including VSD+PS, VSD type, and IVS type), double-outlet right ventricle (DORV; types I–VIII), double-outlet left ventricle (DOLV; IDD/ILL/SDD/SLL), persistent truncus arteriosus (A1–A4, B1–B4), aortopulmonary septal defect (types I–III; including Berry syndrome); aortic valve/aorta—aortic valvar atresia (with VSD/IVS), aortic atresia, aortic stenosis (valvar/supravalvar/subvalvar); aortic arch—arch hypoplasia, interrupted aortic arch (types A/B/C), coarctation (preductal/postductal/ductal type), arch and branch anomalies (isolated branch variants excluding vascular rings, short common trunk, mirror-image branching, etc.), vascular rings (all refined configurations including left/right/double aortic arch, ductal combinations, aberrant branching patterns, etc.); pulmonary arterial system—pulmonary artery sling (partial/complete), pulmonary artery hypoplasia (main trunk; main trunk + branches; branches), unilateral absence of a pulmonary artery (left/right), and anomalous origin

of the pulmonary artery from the ascending aorta (bilateral; left; right proximal; right distal).

The venous return/systemic venous anomalies subgroup includes: anomalous pulmonary venous connection—partial (scimitar syndrome; intracardiac/supracardiac/infracardiac), total (supracardiac/intracardiac/infracardiac/mixed); cor triatriatum—partial (types F/G/H) and complete (types A–E); superior vena cava anomalies—persistent left superior vena cava, absence of right superior vena cava; inferior vena cava anomalies—interruption (including isolated interruption and left atrial isomerism), duplicated inferior vena cava, left-sided inferior vena cava; umbilico-portal-systemic venous anomalies—absence/atresia of the ductus venosus, abnormal ductus venosus connection, portosystemic venous shunts (intrahepatic/extrahepatic), and abnormal umbilical vein connection to the systemic venous system.

The valves/outflow tract subgroup includes: dysplastic valve leaflets; mitral valve—atresia (absent atrioventricular connection/membranous imperforation), dysplasia syndrome, prolapse (anterior leaflet/posterior leaflet/global), stenosis; tricuspid valve—Ebstein anomaly (types A–D), atresia (types I/II/III), dysplasia (types I–III), stenosis; pulmonary stenosis (valvar/supravalvar/subvalvar); pulmonary atresia—PA+VSD (types A/B/C) and PA+IVS (muscular/membranous atresia); tetralogy of Fallot (absent pulmonary valve type/stenotic type); and syndromes related to right ventricular dysplasia.

The physiologic shunt anomalies subgroup includes: foramen ovale—enlarged, premature closure, flow restriction; ductus arteriosus—ductal aneurysm, absence, stenosis, ectopic insertion, tortuous course, increased flow velocity, premature closure, and premature constriction.

The other structural anomalies subgroup includes: single ventricle (single-inlet/double-inlet; indeterminate/mixed/right-ventricular/left-ventricular morphologies and all variants), isolated right ventricular hypoplasia (trabecular segment; trabecular + inlet), hypoplastic left heart syndrome (HLHS: combinations of mitral atresia/stenosis with aortic atresia/stenosis), coronary anomalies—coronary artery fistula (left/right), anomalous origin of a coronary artery from the pulmonary artery (left/right/bilateral), pericardial effusion (localized/diffuse), cardiac protrusive disorders (if incorporated later: atrial/ventricular diverticulum, ventricular aneurysm, etc.), fetal arrhythmia, fetal cardiomyopathy, genetic syndromes, and reduced placental function.

The malposition subgroup includes cardiac positional abnormalities: dextrocardia with situs inversus, mixed dextrocardia, mixed levocardia, ectopia cordis, dextroposition, cardiac dextrorotation, rightward displacement, mesocardia, levorotation, and criss-cross heart.